1) a)

$|\mathbf{a}| = \sqrt{3^2 + 5^2 + (-2)^2} = \sqrt{38}$

1) b)

$|\mathbf{b}| = \sqrt{0^2 + 12^2 + (-16)^2} = \sqrt{400} = 20$

2) a)

$\mathbf{i} \cdot \mathbf{j} = 0$

2) b)

$\mathbf{j} \cdot \mathbf{k} = 0$

2) c)

$\mathbf{i} \cdot \mathbf{i} = 1$

2) d)

$5\mathbf{j} \cdot 3\mathbf{j} = 15$

2) e)

$2\mathbf{j} \cdot 6\mathbf{k} = 0$

2) f)

$\mathbf{j} \cdot (\mathbf{i} + \mathbf{j}) = \mathbf{j} \cdot \mathbf{i} + \mathbf{j} \cdot \mathbf{j} = 0 + 1 = 1$

2) g)

$\mathbf{k} \cdot (2\mathbf{i} - \mathbf{j}) = \mathbf{k} \cdot 2\mathbf{i} - \mathbf{k} \cdot \mathbf{j} = 0 - 0 = 0$

2) h)

$\mathbf{i} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = \mathbf{i} \cdot 3\mathbf{i} - \mathbf{i} \cdot 2\mathbf{j} + \mathbf{i} \cdot \mathbf{k} = 3 - 0 + 0 = 3$

2) i)

$(\mathbf{i} + \mathbf{j}) \cdot (\mathbf{i} + \mathbf{k}) = \mathbf{i} \cdot \mathbf{i} + \mathbf{i} \cdot \mathbf{k} + \mathbf{j} \cdot \mathbf{i} + \mathbf{j} \cdot \mathbf{k} = 1 + 0 + 0 + 0 = 1$

2) j)

$(2\mathbf{i} - 3\mathbf{k}) \cdot (2\mathbf{j} - 3\mathbf{k}) = 2\mathbf{i} \cdot 2\mathbf{j} - 2\mathbf{i} \cdot 3\mathbf{k} - 3\mathbf{k} \cdot 2\mathbf{j} + 3\mathbf{k} \cdot 3\mathbf{k} = 0 - 0 + 0 + 9 = 9$

2) k)

$(\mathbf{i}-\mathbf{j}-\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}-3\mathbf{k})$
$=\mathbf{i}\cdot\mathbf{i}+\mathbf{i}\cdot 2\mathbf{j}-\mathbf{i}\cdot 3\mathbf{k}-\mathbf{j}\cdot 2\mathbf{j}+\mathbf{j}\cdot 3\mathbf{k}-\mathbf{k}\cdot\mathbf{i}-\mathbf{k}\cdot 2\mathbf{j}+\mathbf{k}\cdot 3\mathbf{k}$
$=1+0-0-2+0-0-0+3=2$

3) a)

$\mathbf{a}\cdot\mathbf{b}=4\times 10+2\times 20+3\times(-5)=65$

3) b)

$\mathbf{a}\cdot\mathbf{b}=6\times(-3)+1\times 2+(-1)\times(-5)=-11$

3) c)

$\mathbf{a}\cdot\mathbf{b}=1\times 5+(-1)\times 2+3\times(-1)=0$

3) d)

$\mathbf{a}\cdot\mathbf{b}=8\times 16+0\times(-18)+(-15)\times 0=128$

4)

$(2\mathbf{i}+p\mathbf{j}-3p\mathbf{k})\cdot(3p\mathbf{i}-2\mathbf{j}-\mathbf{k})=2\times 3p+p\times(-2)+(-3p)\times(-1)$
$=6p-2p+3p=7p$

5)

$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{1\times(-1)+1\times 3+0\times 0}{\sqrt{1^2+1^2+0^2}\times\sqrt{(-1)^2+3^2+0^2}}=\frac{2}{\sqrt{2}\sqrt{10}}=\frac{\sqrt{5}}{5}$

6) a)

$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{1\times 2+3\times 4+4\times 6}{\sqrt{1^2+3^2+4^2}\times\sqrt{2^2+4^2+6^2}}=\frac{38}{\sqrt{26}\sqrt{56}}=\frac{19\sqrt{91}}{182}$
$\theta=\arccos\left(\frac{19\sqrt{91}}{182}\right)=5.2°$ (1 DP)

6) b)

$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{3\times 5+2\times(-1)+(-3)\times 2}{\sqrt{3^2+2^2+(-3)^2}\times\sqrt{5^2+(-1)^2+2^2}}=\frac{7}{\sqrt{22}\sqrt{30}}=\frac{7\sqrt{165}}{330}$
$\theta=\arccos\left(\frac{7\sqrt{165}}{330}\right)=74.2°$ (1 DP)

6) c)

$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{\sqrt{2}\times(-3)+3\times 5\sqrt{2}+2\sqrt{2}\times 5}{\sqrt{(\sqrt{2})^2+3^2+(2\sqrt{2})^2}\times\sqrt{(-3)^2+(5\sqrt{2})^2+5^2}}=\frac{22\sqrt{2}}{\sqrt{19}\sqrt{84}}=\frac{11\sqrt{798}}{399}$
$\theta=\arccos\left(\frac{11\sqrt{798}}{399}\right)=38.8°$ (1 DP)

6) d)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\times\left(-\frac{1}{\sqrt{2}}\right)+(-1)\times\frac{1}{\sqrt{2}}}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2+\left(\frac{1}{\sqrt{2}}\right)^2+(-1)^2}\times\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2+\left(-\frac{1}{\sqrt{2}}\right)^2+\left(\frac{1}{\sqrt{2}}\right)^2}} = \frac{-\frac{1}{\sqrt{2}}}{\sqrt{2}\sqrt{\frac{3}{2}}} = \frac{-\frac{\sqrt{2}}{2}}{\sqrt{2}\sqrt{\frac{3}{2}}} = -\frac{\sqrt{6}}{6}$$

$$\theta = \arccos\left(-\frac{\sqrt{6}}{6}\right) = 114.094\ldots°$$

Acute angle $= 180° - 114.094\ldots° = 65.9°$ (1 DP)

7)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \Rightarrow \cos\theta = \frac{8}{2\times\frac{11}{2}} \Rightarrow \cos\theta = \frac{8}{2\times\frac{11}{2}} = \frac{8}{11}$$

$$\therefore \sin\theta = \frac{\sqrt{57}}{11}$$

8)

$$\overrightarrow{OA} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k} \Rightarrow \overrightarrow{AO} = -2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$$

$$\overrightarrow{OB} = \mathbf{i} - \mathbf{j} - \mathbf{k}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}$$

$$\cos\theta = \frac{\overrightarrow{AO}\cdot\overrightarrow{AB}}{|\overrightarrow{AO}||\overrightarrow{AB}|} = \frac{(-2)\times(-1)+3\times2+(-6)\times(-7)}{\sqrt{(-2)^2+3^2+(-6)^2}\times\sqrt{(-1)^2+2^2+(-7)^2}} = \frac{50}{\sqrt{49}\sqrt{54}} = \frac{25\sqrt{6}}{63}$$

$$\theta = \arccos\left(\frac{25\sqrt{6}}{63}\right) = 13.6° \text{ (1 DP)}$$

9) a)

$$\overrightarrow{OA} = -4\mathbf{i} + \mathbf{j} - 8\mathbf{k}$$

$$\overrightarrow{OB} = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$$

$$\cos\theta = \frac{\overrightarrow{OA}\cdot\overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|} = \frac{(-4)\times2+1\times4+(-8)\times(-4)}{\sqrt{(-4)^2+1^2+(-8)^2}\times\sqrt{2^2+4^2+(-4)^2}} = \frac{28}{\sqrt{81}\sqrt{36}} = \frac{14}{27}$$

9) b)

$$\sin\theta = \frac{\sqrt{533}}{27}$$

Area $= \frac{1}{2}\left|\overrightarrow{OA}\right|\left|\overrightarrow{OB}\right|\sin\theta = \frac{1}{2}\times\sqrt{81}\times\sqrt{36}\times\frac{\sqrt{533}}{27} = \sqrt{533}$ units^2

10)

$\overrightarrow{OA} = -3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

$\overrightarrow{OB} = 8\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$

$\overrightarrow{OC} = 2\mathbf{i} - 9\mathbf{j} - 10\mathbf{k}$

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = -11\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = -6\mathbf{i} - 5\mathbf{j} - 14\mathbf{k}$

$\cos\theta = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{(-11)\times(-6)+3\times(-5)+(-2)\times(-14)}{\sqrt{(-11)^2+3^2+(-2)^2}\times\sqrt{(-6)^2+(-5)^2+(-14)^2}} = \frac{79}{\sqrt{134}\sqrt{257}} = \frac{79\sqrt{34438}}{34438}$

$\theta = \arccos\left(\frac{79\sqrt{34438}}{34438}\right) = 64.8°$ (1 DP)

11) a)

$\mathbf{a}\cdot\mathbf{b} = 6\times 4 + (-1)\times 4 + 4\times(-5) = 0$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular

11) b)

$\mathbf{a}\cdot\mathbf{b} = (-2)\times 3 + 4\times(-6) + 5\times(-6) = -60 \neq 0$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are not perpendicular

11) c)

$\mathbf{a}\cdot\mathbf{b} = 9\times(-5) + (-6)\times(-10) + 15\times(-1) = 0$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular

12) a)

$\overrightarrow{OA} = 8\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

$\overrightarrow{OB} = 10\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$

$\overrightarrow{OC} = 10\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

$\cos\theta = \frac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{|\overrightarrow{AB}||\overrightarrow{AC}|} = \frac{2\times2+(-4)\times(-2)+4\times1}{\sqrt{2^2+(-4)^2+4^2}\times\sqrt{2^2+(-2)^2+1^2}} = \frac{16}{\sqrt{36}\sqrt{9}} = \frac{8}{9}$

$\therefore \sin\theta = \frac{\sqrt{17}}{9}$

12) b)

$\overrightarrow{OD} = 12\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$

$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = 4\mathbf{i} + 6\mathbf{j} + 4\mathbf{k}$

$\overrightarrow{AD}\cdot\overrightarrow{AB} = 4\times 2 + 6\times(-4) + 4\times 4 = 0$

$\overrightarrow{AD}\cdot\overrightarrow{AC} = 4\times 2 + 6\times(-2) + 4\times 1 = 0$

$\therefore$ $\overrightarrow{AD}$ is perpendicular to $\overrightarrow{AB}$ and $\overrightarrow{AC}$

12) c)

Area of Triangle $ABC = \frac{1}{2}\left|\overrightarrow{AB}\right|\left|\overrightarrow{AC}\right|\sin\theta = \frac{1}{2} \times \sqrt{36} \times \sqrt{9} \times \frac{\sqrt{17}}{9} = \sqrt{17}$

$\left|\overrightarrow{AD}\right| = \sqrt{4^2 + 6^2 + 4^2} = \sqrt{68} = 2\sqrt{17}$

Volume of Pyramid $= \frac{1}{3} \times \sqrt{17} \times 2\sqrt{17} = \frac{34}{3}$ units^3

13) a)

$\overrightarrow{OA} = \mathbf{i} + 9\mathbf{j} - 6\mathbf{k}$

$\overrightarrow{OB} = 2\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$

$\overrightarrow{OC} = \frac{11}{3}\mathbf{i} + \frac{28}{3}\mathbf{j} + \frac{7}{12}\mathbf{k}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{i} - 4\mathbf{j} + 8\mathbf{k}$

$\overrightarrow{OD} = \overrightarrow{OA} + \frac{2}{3}\overrightarrow{AB} = (\mathbf{i} + 9\mathbf{j} - 6\mathbf{k}) + \frac{2}{3}(\mathbf{i} - 4\mathbf{j} + 8\mathbf{k})$

$\therefore \overrightarrow{OD} = \frac{5}{3}\mathbf{i} + \frac{19}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

$\overrightarrow{DC} = \overrightarrow{OC} - \overrightarrow{OD} = 2\mathbf{i} + 3\mathbf{j} + \frac{5}{4}\mathbf{k}$

$\overrightarrow{AB} \cdot \overrightarrow{DC} = 1 \times 2 + (-4) \times 3 + 8 \times \frac{5}{4} = 0$

$\therefore \overrightarrow{AB}$ and $\overrightarrow{DC}$ are perpendicular

13) b)

$\left|\overrightarrow{AB}\right| = \sqrt{1^2 + (-4)^2 + 8^2} = \sqrt{81} = 9$

$\left|\overrightarrow{DC}\right| = \sqrt{2^2 + 3^2 + \left(\frac{5}{4}\right)^2} = \sqrt{\frac{233}{16}} = \frac{\sqrt{233}}{4}$

Area of triangle $ABC = \frac{1}{2}\left|\overrightarrow{AB}\right|\left|\overrightarrow{DC}\right| = \frac{1}{2} \times 9 \times \frac{\sqrt{233}}{4} = \frac{9\sqrt{233}}{8}$ units^2

14) a)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 3p \times 2 + 2 \times (-4p) + (-1) \times 6 = 0$

$\Rightarrow 6p - 8p - 6 = 0 \Rightarrow -2p - 6 = 0 \Rightarrow p = -3$

14) b)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 16 \times (-3) + (-4p) \times 7 + 7 \times 40p = 0$

$\Rightarrow -48 - 28p + 280p = 0 \Rightarrow -48 + 252p = 0 \Rightarrow p = \frac{4}{21}$

14) c)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 5p \times p + (-p) \times (-5) + 1 \times 8p = 0$

$\Rightarrow 5p^2 + 5p + 8p = 0 \Rightarrow 5p^2 + 13p = 0 \Rightarrow p(5p + 13) = 0 \Rightarrow p = 0$ or $p = -\frac{13}{5}$

14) d)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 4p \times (-3p) + (-6p) \times 2 + 12 \times 6 = 0$

$\Rightarrow -12p^2 - 12p + 72 = 0 \Rightarrow p^2 + p - 6 = 0 \Rightarrow (p+3)(p-2) = 0 \Rightarrow p = -3$ or $p = 2$

15) a)

$\overrightarrow{OA} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$

$\overrightarrow{OB} = \mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$

$\overrightarrow{OC} = 6\mathbf{i} + (p-5)\mathbf{j} - \mathbf{k}$

$\overrightarrow{CA} = \overrightarrow{OA} - \overrightarrow{OC} = -4\mathbf{i} + (4-p)\mathbf{j} + 4\mathbf{k}$

$\overrightarrow{CB} = \overrightarrow{OB} - \overrightarrow{OC} = -5\mathbf{i} + (10-p)\mathbf{j} + 5\mathbf{k}$

Angle $ACB = 90° \Rightarrow \overrightarrow{CA} \cdot \overrightarrow{CB} = 0$

$\therefore (-4) \times (-5) + (4-p) \times (10-p) + 4 \times 5 = 0$

$\Rightarrow 20 + 40 - 14p + p^2 + 20 = 0 \Rightarrow p^2 - 14p + 80 = 0$

$b^2 - 4ac = (-14)^2 - 4 \times 1 \times 80 = -124 < 0$

So angle ACB can never be 90°

15) b) i)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \mathbf{i} - 6\mathbf{j} - \mathbf{k}$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = 5\mathbf{i} + (p-10)\mathbf{j} - 5\mathbf{k}$

Angle $ABC = 90° \Rightarrow \overrightarrow{BA} \cdot \overrightarrow{BC} = 0$

$\therefore 1 \times 5 + (-6) \times (p-10) + (-1) \times (-5) = 0$

$\Rightarrow 5 - 6p + 60 + 5 = 0 \Rightarrow 70 - 6p \Rightarrow p = \frac{35}{3}$

15) b) ii)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -\mathbf{i} + 6\mathbf{j} + \mathbf{k}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 4\mathbf{i} + (p-4)\mathbf{j} - 4\mathbf{k}$

If A, B and C are collinear then $\overrightarrow{AB} = k\overrightarrow{AC}$ for some constant k

$-1 = 4k \Rightarrow k = -\frac{1}{4}$

$6 = (p-4)k$

$1 = -4k \Rightarrow k = -\frac{1}{4}$

$\therefore 6 = (p-4) \times -\frac{1}{4} \Rightarrow p = -20$

16)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow \begin{pmatrix} 3k \\ -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix} = 0 \Rightarrow (3k) \times 4 + (-2) \times (-3) + 5 \times (-2) = 0$

$\Rightarrow 12k + 6 - 10 = 0 \Rightarrow k = \frac{1}{3}$

17)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow \begin{pmatrix} -6 \\ 2(k+1)^2 \\ 29 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -5 \\ 2 \end{pmatrix} = 0 \Rightarrow (-6) \times 8 + 2(k+1)^2 \times (-5) + 29 \times 2 = 0$

$\Rightarrow -48 - 10(k+1)^2 + 58 = 0 \Rightarrow (k+1)^2 = 1 \Rightarrow k + 1 = \pm 1 \Rightarrow k = -2$ or $k = 0$

18)

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2 \times 1 + 3 \times m + 5 \times n = 0 \Rightarrow 3m + 5n = -2$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 5 \times 1 + 4 \times m + 2 \times n = 0 \Rightarrow 4m + 2n = -5$ (2)

$\therefore m = -\frac{3}{2}$ and $n = \frac{1}{2}$

19)

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow \begin{pmatrix} -13 \\ -4 \\ -9 \end{pmatrix} \cdot \begin{pmatrix} m \\ n \\ 5 \end{pmatrix} = 0 \Rightarrow (-13) \times m + (-4) \times n + (-9) \times 5 = 0$

$\Rightarrow -13m - 4n = 45$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow \begin{pmatrix} 3 \\ -7 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} m \\ n \\ 5 \end{pmatrix} = 0 \Rightarrow 3 \times m + (-7) \times n + 10 \times 5 = 0$

$\Rightarrow 3m - 7n = -50$ (2)

$\therefore m = -5$ and $n = 5$

20)

Let $\mathbf{c} = \mathbf{i} + m\mathbf{j} + n\mathbf{k}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 1 \times 1 + (-2) \times m + 3 \times n = 0 \Rightarrow -2m + 3n = -1$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 4 \times 1 + (-5) \times m + 2 \times n = 0 \Rightarrow -5m + 2n = -4$ (2)

$\therefore m = \frac{10}{11}$ and $n = \frac{3}{11}$

So $\mathbf{c} = \mathbf{i} + \frac{10}{11}\mathbf{j} + \frac{3}{11}\mathbf{k}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

21)

Let $\mathbf{c} = \mathbf{i} + m\mathbf{j} + n\mathbf{k}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow (-42) \times 1 + (-1) \times m + 6 \times n = 0 \Rightarrow -m + 6n = 42$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 10 \times 1 + 5 \times m + (-5) \times n = 0 \Rightarrow 5m - 5n = -10$ (2)

$\therefore m = 6$ and $n = 8$

So $\mathbf{c} = \mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

$\hat{\mathbf{c}} = \frac{\mathbf{c}}{|\mathbf{c}|} = \frac{1}{\sqrt{1^2+6^2+8^2}}(\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}) = \frac{1}{\sqrt{101}}(\mathbf{i} + 6\mathbf{j} + 8\mathbf{k})$

$= \frac{1}{\sqrt{101}}\mathbf{i} + \frac{6}{\sqrt{101}}\mathbf{j} + \frac{8}{\sqrt{101}}\mathbf{k}$ is a unit vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

22)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow p \times 1 + 3p \times m + 5 \times n = 0 \Rightarrow p + 3pm + 5n = 0$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow (-2) \times 1 + 4 \times m + 3 \times n = 0 \Rightarrow -2 + 4m + 3n = 0$ (2)

(1)$\times$ 3: $3p + 9pm + 15n = 0$ (3)

(2)$\times$ 5: $-10 + 20m + 15n = 0$ (4)

(3)-(4): $3p + 9pm + 10 - 20m = 0 \Rightarrow 3p + 10 = (20 - 9p)m \Rightarrow m = \frac{3p+10}{20-9p}$

$\therefore -2 + 4 \times \frac{3p+10}{20-9p} + 3n = 0 \Rightarrow n = \frac{1}{3}\left(2 - \frac{12p+40}{20-9p}\right) = \frac{1}{3}\left(\frac{40-18p}{20-9p} - \frac{12p+40}{20-9p}\right)$

$= \frac{1}{3}\left(\frac{-30p}{20-9p}\right) = -\frac{10p}{20-9p}$

So $\mathbf{c} = \begin{pmatrix} 1 \\ \frac{3p+10}{20-9p} \\ -\frac{10p}{20-9p} \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

An alternative is $\begin{pmatrix} 20-9p \\ 3p+10 \\ -10p \end{pmatrix}$

23)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2q \times 1 + (-5) \times m + 3q \times n = 0 \Rightarrow 2q - 5m + 3qn = 0$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 5 \times 1 + 7 \times m + 1 \times n = 0 \Rightarrow 5 + 7m + n = 0$ (2)

(1)× 7: $14q - 35m + 21qn = 0$ (3)

(2)× 5: $25 + 35m + 5n = 0$ (4)

(3)+(4): $14q + 21qn + 25 + 5n = 0 \Rightarrow (21q + 5)n = -(14q + 25) \Rightarrow n = -\frac{14q+25}{21q+5}$

$\therefore 5 + 7m - \frac{14q+25}{21q+5} = 0 \Rightarrow m = \frac{1}{7}\left(\frac{14q+25}{21q+5} - 5\right) = \frac{1}{7}\left(\frac{14q+25}{21q+5} - \frac{105q+25}{21q+5}\right)$

$= \frac{1}{7}\left(-\frac{91q}{21q+5}\right) = -\frac{13q}{21q+5}$

So $\mathbf{c} = \begin{pmatrix} 1 \\ -\frac{13q}{21q+5} \\ -\frac{14q+25}{21q+5} \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

An alternative is $\begin{pmatrix} 21q + 5 \\ -13q \\ -14q - 25 \end{pmatrix}$

24)

$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} x+1 \\ x-2 \\ 2x+3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 4x \\ 7 \end{pmatrix} = (x+1) \times (-2) + (x-2) \times 4x + (2x+3) \times 7$

$= -2x - 2 + 4x^2 - 8x + 14x + 21$

$= 4x^2 + 4x + 19 = 4(x^2 + x) + 19 = 4\left(\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}\right) + 19 = 4\left(x + \frac{1}{2}\right)^2 - 1 + 19$

$= 4\left(x + \frac{1}{2}\right)^2 + 18 > 0$ for all real values of x, so $\mathbf{a} \cdot \mathbf{b} \neq 0$ and so $\mathbf{a}$ and $\mathbf{b}$ can never be perpendicular.

25)

$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} 3-x \\ 4-3x \\ 5-x \end{pmatrix} \cdot \begin{pmatrix} x-1 \\ x \\ -5 \end{pmatrix} = (3-x) \times (x-1) + (4-3x) \times x + (5-x) \times (-5)$

$= 3x - 3 - x^2 + x + 4x - 3x^2 - 25 + 5x$

$= -4x^2 + 13x - 28 = -4\left(x^2 - \frac{13}{4}x\right) - 28 = -4\left(\left(x - \frac{13}{8}\right)^2 - \frac{169}{64}\right) - 28$

$= -4\left(x - \frac{13}{8}\right)^2 + \frac{169}{16} - 28 = -4\left(x - \frac{13}{8}\right)^2 - \frac{279}{16} < 0$ for all real values of x, so $\mathbf{a} \cdot \mathbf{b} \neq 0$ and so $\mathbf{a}$ and $\mathbf{b}$ can never be perpendicular.

26) a)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2 \times 1 + q \times m + (-3) \times n = 0 \Rightarrow 2 + mq - 3n = 0$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow (q+1) \times 1 + q \times m + (-1) \times n = 0 \Rightarrow q + 1 + mq - n = 0$ (2)

(2)-(1): $q - 1 + 2n = 0 \Rightarrow n = \frac{1}{2}(1-q)$

$\therefore q + 1 + mq - \frac{1}{2}(1-q) = 0 \Rightarrow q + 1 + mq - \frac{1}{2} + \frac{1}{2}q \Rightarrow \frac{3}{2}q + \frac{1}{2} + mq = 0$

$\Rightarrow m = -\frac{1}{q}\left(\frac{3}{2}q + \frac{1}{2}\right) \Rightarrow m = -\frac{3q+1}{2q}$

So $\mathbf{c} = \begin{pmatrix} 1 \\ -\frac{3q+1}{2q} \\ \frac{1}{2}(1-q) \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

An alternative is $\begin{pmatrix} 2q \\ -3q-1 \\ q-q^2 \end{pmatrix}$

26) b)

$\mathbf{c} \cdot \mathbf{d} = 2q \times (-5q) + (-3q-1) \times 5 + (q-q^2) \times 1 = -10q^2 - 15q - 5 + q - q^2$

$= -11q^2 - 14q - 5 = -11\left(q^2 + \frac{14}{11}q\right) - 5 = -11\left(\left(q + \frac{7}{11}\right)^2 - \frac{49}{121}\right) - 5$

$= -11\left(q + \frac{7}{11}\right)^2 + \frac{49}{11} - 5 = -11\left(q + \frac{7}{11}\right)^2 - \frac{6}{11} < 0$ for all real values of q, so $\mathbf{c} \cdot \mathbf{d} \neq 0$ and so $\mathbf{c}$ and $\mathbf{d}$ can never be perpendicular.

27)

Let $\mathbf{c} = \begin{pmatrix} m \\ n \\ 1 \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2 \times m + (-3) \times n + x \times 1 = 0 \Rightarrow 2m - 3n + x = 0$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow (-4) \times m + 3 \times n + (x + 5) \times 1 = 0 \Rightarrow -4m + 3n + x + 5 = 0$ (2)

(1)+(2): $-2m + 2x + 5 = 0 \Rightarrow m = \frac{1}{2}(2x + 5)$

$\therefore 2 \times \frac{1}{2}(2x + 5) - 3n + x = 0 \Rightarrow n = \frac{1}{3}\left(2 \times \frac{1}{2}(2x + 5) + x\right) = \frac{1}{3}(3x + 5)$

So $\mathbf{c} = \begin{pmatrix} \frac{1}{2}(2x + 5) \\ \frac{1}{3}(3x + 5) \\ 1 \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

An alternative is $\begin{pmatrix} 6x + 15 \\ 6x + 10 \\ 6 \end{pmatrix}$

To find the shortest possible vector, we need to minimise

$(6x + 15)^2 + (6x + 10)^2 + 6^2 = 36x^2 + 180x + 225 + 36x^2 + 120x + 100 + 36$

$= 72x^2 + 300x + 361 = 72\left(x^2 + \frac{25}{6}x\right) + 361 = 72\left(\left(x + \frac{25}{12}\right)^2 - \frac{625}{144}\right) + 361$

$= 72\left(x + \frac{25}{12}\right)^2 - \frac{625}{2} + 361 = 72\left(x + \frac{25}{12}\right)^2 + \frac{97}{2}$

This is minimised when $x = -\frac{25}{12}$

So $\mathbf{c} = \begin{pmatrix} \frac{5}{12} \\ -\frac{5}{12} \\ 1 \end{pmatrix}$ is the shortest possible vector of the required form that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

28)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$

$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 1 \times 1 + x \times m + 4 \times n = 0 \Rightarrow 1 + mx + 4n = 0$ (1)

$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 5 \times 1 + (1 - x) \times m + 8 \times n = 0 \Rightarrow 5 + m - mx + 8n = 0$ (2)

(1)× 2: $2 + 2mx + 8n = 0$ (3)

(3)-(2): $-3 + 3mx - m = 0 \Rightarrow (3x - 1)m = 3 \Rightarrow m = \frac{3}{3x-1}$

$\therefore 1 + \frac{3}{3x-1} \times x + 4n = 0 \Rightarrow n = \frac{1}{4}\left(-\frac{3x}{3x-1} - 1\right) = -\frac{1}{4}\left(\frac{3x}{3x-1} + 1\right) = -\frac{1}{4}\left(\frac{3x}{3x-1} + \frac{3x-1}{3x-1}\right)$

$= -\frac{1}{4}\left(\frac{6x-1}{3x-1}\right)$

So $\mathbf{c} = \begin{pmatrix} 1 \\ \frac{3}{3x-1} \\ -\frac{1}{4}\left(\frac{6x-1}{3x-1}\right) \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

An alternative is $\begin{pmatrix} 3x - 1 \\ 3 \\ -\frac{1}{4}(6x - 1) \end{pmatrix}$ or $\begin{pmatrix} 4 - 12x \\ -12 \\ 6x - 1 \end{pmatrix}$

To find the shortest possible vector, we need to minimise

$(4 - 12x)^2 + (-12)^2 + (6x - 1)^2 = 144x^2 - 96x + 16 + 144 + 36x^2 - 12x + 1$

$= 180x^2 - 108x + 161 = 180\left(x^2 - \frac{3}{5}x\right) + 161 = 180\left(\left(x - \frac{3}{10}\right)^2 - \frac{9}{100}\right) + 161$

$= 180\left(x - \frac{3}{10}\right)^2 - \frac{81}{5} + 161 = 180\left(x - \frac{3}{10}\right)^2 + \frac{724}{5}$

This is minimised when $x = \frac{3}{10}$

So $\mathbf{c} = \begin{pmatrix} 1 \\ -30 \\ 2 \end{pmatrix}$ is the shortest possible vector of the required form that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

29)

Let $\mathbf{p} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where not all of a, b, c are zero, and $\mathbf{q} = \begin{pmatrix} d \\ e \\ f \end{pmatrix}$, where not all of d, e, f are zero.

$$\mathbf{p} + \mathbf{q} = \begin{pmatrix} a + d \\ b + e \\ c + f \end{pmatrix}$$

$$\mathbf{p} - \mathbf{q} = \begin{pmatrix} a - d \\ b - e \\ c - f \end{pmatrix}$$

If $\mathbf{p} + \mathbf{q}$ is perpendicular to $\mathbf{p} - \mathbf{q}$ then $(\mathbf{p} + \mathbf{q}) \cdot (\mathbf{p} - \mathbf{q}) = 0$

$$\begin{pmatrix} a + d \\ b + e \\ c + f \end{pmatrix} \cdot \begin{pmatrix} a - d \\ b - e \\ c - f \end{pmatrix} = 0$$

$$\Rightarrow (a + d)(a - d) + (b + e)(b - e) + (c + f)(c - f) = 0$$

$$\Rightarrow a^2 - d^2 + b^2 - e^2 + c^2 - f^2 = 0$$

$$\Rightarrow a^2 + b^2 + c^2 = d^2 + e^2 + f^2$$

$$\Rightarrow \sqrt{a^2 + b^2 + c^2} = \sqrt{d^2 + e^2 + f^2}$$

$$\Rightarrow |\mathbf{p}| = |\mathbf{q}|$$